

Characterization of High Speed Phase Frequency Detector Circuit

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Abstract

In this paper the performance of high speed phase frequency detector (PFD) is analysed. A high-speed PFD reduces the power consumption of traditional PFD to 0.423 nW at 1 GHz clock frequency and dead zone to 2 ps. It is suitable for low power applications. A PFD uses only 18 transistors. This PFD operates up to 1 GHz at 1.2 V supply voltage. The proposed architecture of the PFD has been implemented using 0.35 μ m CMOS technology and 90 nm CMOS technology.

Keywords: low power PFD, phase frequency detector, phase-locked loop, PLL

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INTRODUCTION

The Phase Frequency Detector (PFD) is the heart of the Phase Locked Loops(PLL). As a consequence, many techniques of implementation of PFD have been introduced to minimize the power consumption and reduction in Dead Zone of new VLSI circuits [1]. This paper is on the design and analysis of PFD is an endeavour in that direction.

PHASE DETECTOR (PD)

A phase detector (PD) is the circuit which compares the phase of two input signals. PD circuit has two inputs, usually one from a voltage-controlled oscillator (VCO) and another from some external reference source. It has two output signals to adjust to lock onto the phase and frequency (Figures 1 and 2). The PFD is an improved circuit over the phase comparators of phase detectors of early PLLs. PLL can provide a frequency error output and a phase error signal [1–3, 9].

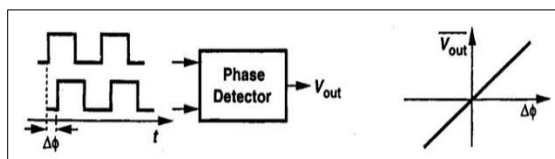


Fig. 1: Characteristic of an Ideal PD[6].

PLL can be formed as follows: The PFD phase/frequency error output is the input of a loop filter which integrates the signal. The output of loop filter is fed back to VCO. VCO

generates an output signal with a frequency that is proportional to the input voltage. The VCO output is also fed back to the PFD to design the PLL[4, 5, 9].

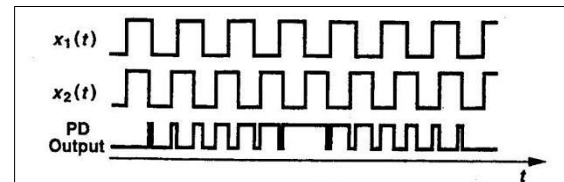


Fig. 2: Input and Output Waveforms of a PD[5].

PHASE FREQUENCY DETECTOR (PFD)

The multiplier and XOR are the sequential circuits that produce two outputs which are not compliment to each other. Such circuits are referred as sequential phase/frequency detectors. The typical PFD operation is explained in Figure 3 [11, 13].

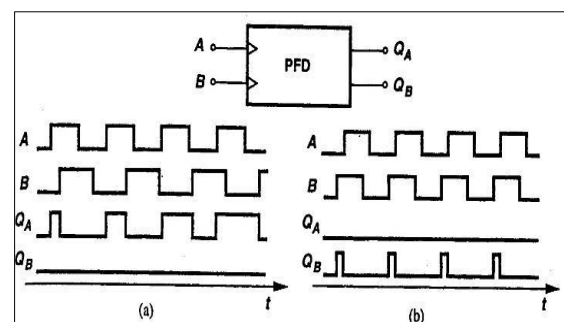


Fig. 3: Operation of a Typical PFD[13].

The operation of PFD: The PFD produces the positive pulses at either at Q_A or Q_B . If the frequency at input A is greater than at input B, then positive pulses are produced at Q_A , and Q_B remains zero. On the same frequency at input A and B, the Q_A and Q_B both remain zero. The width of the pulses depends on the phase difference of the two inputs. Such outputs Q_A and Q_B are also known as Up and Down signals respectively [13]. This basic Phase Frequency Detector is also known as “three states PFD”. It can be explained also as a three-state machine.

Table 1: Three-state PFD[9].

	State 0	State 1	State 2	No State Allowed
Q_A (UP)	0	1	0	1
Q_B (DOWN)	0	0	1	1

The states can be changed by the rising transition of inputs A and B of the circuit. The state diagram of PFD is shown in Figure 4. Suppose PFD is in state zero, the transition on input A takes into state I. But if the transition occurs on input B, then the PFD return to state 0. Similar function can exist between state 0 and II [9].

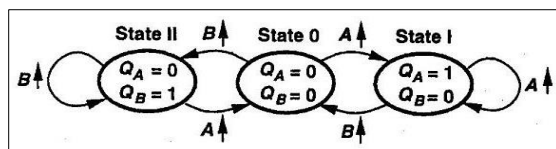


Fig. 4: State diagram of a PFD[9].

The AND-based PFD is formed as shown in Figure 5. There are two D-FF (flip flops) and an AND gate. When the clock is applied at the inputs D flip flops, they are activated where the input PFD signals are reference signal and VCO. As the one input D of the FF is ‘1’, the output Q will be high at the positive rising edge of the clock input.

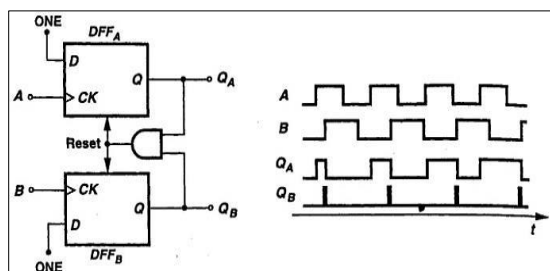


Fig. 5: Implementation of PFD[1].

Both D-FF will be reset by the reset signal which is generated through the AND gate when two outputs of D-FF are high. At instant the outputs become down again. These signals are UP and Down. The range of clock is normally in MHz of such design. Figure 6 shows the input-output characteristic of the PFD.

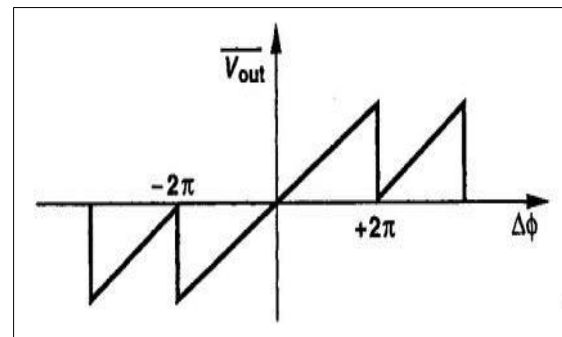


Fig. 6: PFD Characteristic[7].

PROBLEMS IN THE CONVENTIONAL PFD

The high speed of PFD can be achieved by reducing the pulse width and propagation delay of reset signal. In AND-based PFD, the reset signal is affected by delay of D-FF. In such design, the range of the operating frequency is 1 GHz. The disadvantage of such PFD is dead-zone. The dead zone is the small difference in the phase of the inputs that a PFD will not be able to detect. Due to this dead zone, the jitter will be produced at the output. General PFD has very high dead zone, thus the jitter [6,7].

HIGH SPEED PHASE FREQUENCY DETECTOR

The High Speed Phase Frequency Detector circuit is discussed here which detects the phase and frequency difference between two inputs. The circuits detect both the edges of the PFD and so it is called the high speed PFD [2, 3].

The Phase Frequency Detector circuit has two inputs, one the clock reference (CLKREF) and the other is the clock from voltage control oscillator (CLKVCO). Two outputs are UP signal and DOWN signal. The outputs are connected to the charge pump to generate the related control signal for VCO. Input to the PFD is from the reference clock.

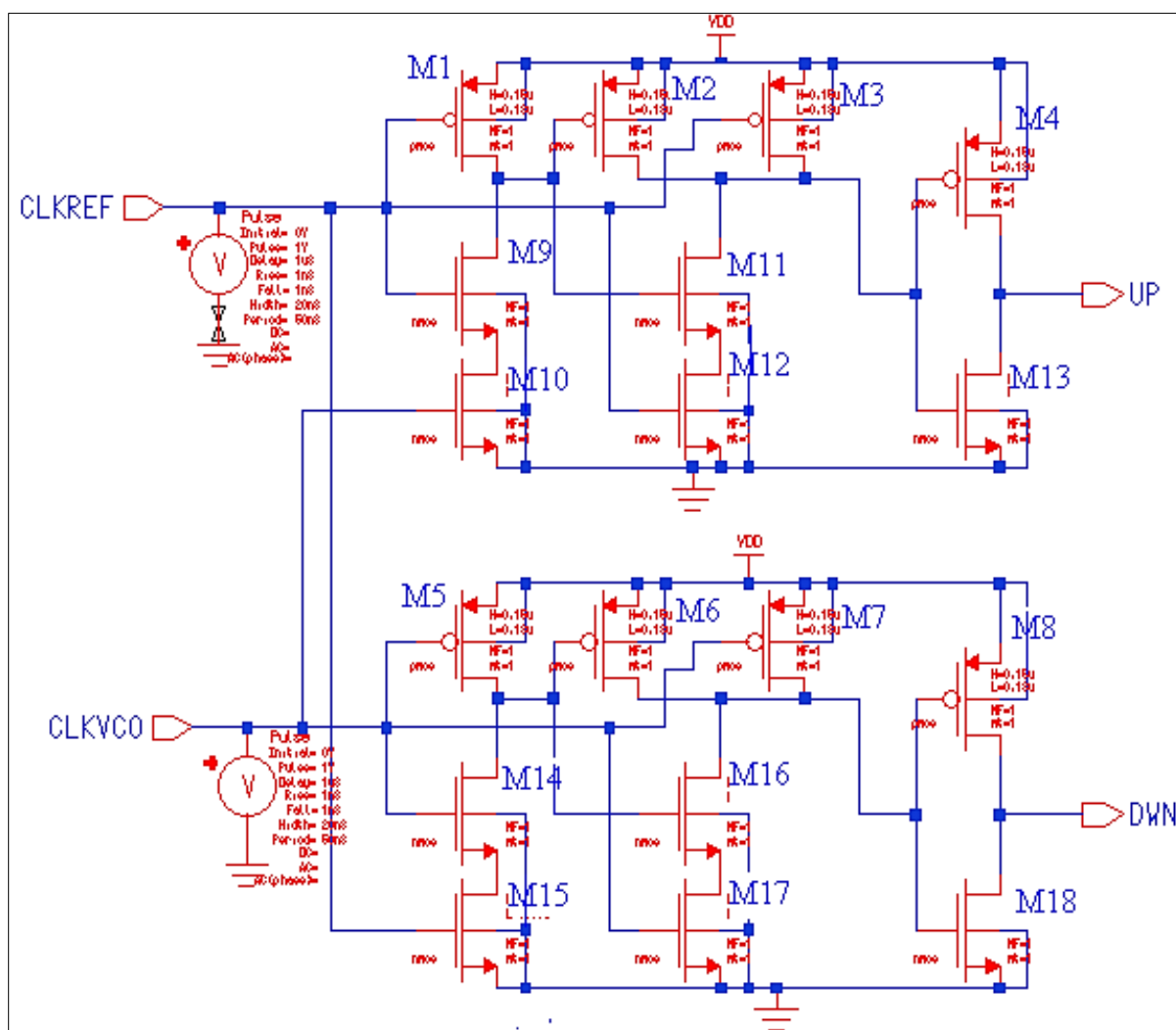


Fig. 7: High Speed PFD[3, 11, 12, 13].

The power consumption of the traditional PFD is quite high. So, as a solution in this paper, high speed PFD is used with lower jitter and power. In the high speed PFD, the feedback path is totally eliminated compared to traditional PFD. So improvement is possible with high speed PFD which is another way to reduce the power consumption and the jitter bug addition of two invertors and NAND gates.

The implementation of the high speed PFD with CMOS .35 μ m technology is as shown in the Figure 7.

TRANSIENT ANALYSIS OF HIGH SPEED PFD

The different types Phase Frequency Detector suitable for high speed applications can be characterized and simulated. The Phase

Frequency Detector as shown in Figure 7 is designed in TSMC 0.35 μ m and in Generic 90 nm technologies. When the CLKREF signal is leading, then the Up output signal results high and when CLKVCO signal is leading then it will result DOWN signal high. Figure 8 A shows the simulated output of High Speed PFD at 50 MHz input frequency. Figure 8 shows both the input signals–data and the clock are at same time so, both the output signals–UPsignal and DOWN signal are low.

Figure 8 A-B-C shows the simulation results in 90 nm CMOS technology of proposed high speed phase frequency detector circuit. Figure 9 A-B-C shows the simulation results in 0.35 μ m CMOS technology of proposed high speed phase frequency detector circuit. Figure 8-A shows the lock condition of the High Speed PFD at 20ns and the power dissipation is

reduced compare to traditional PFD. Figure 8-C shows the Dead Zone of the High Speed PFD. With the use of modification in the reset path and the circuit of the D flip-flop power dissipation is reduced. With this high-speed

circuit of PFD, the power dissipation is reduced and the dead zone is also reduced. In this type of high-speed PFD, 2 ps of dead zone is achieved at 1 GHz. So, the output jitter is reduced with this circuit of High Speed PFD.

SIMULATION RESULTS IN 90 nm TECHNOLOGY

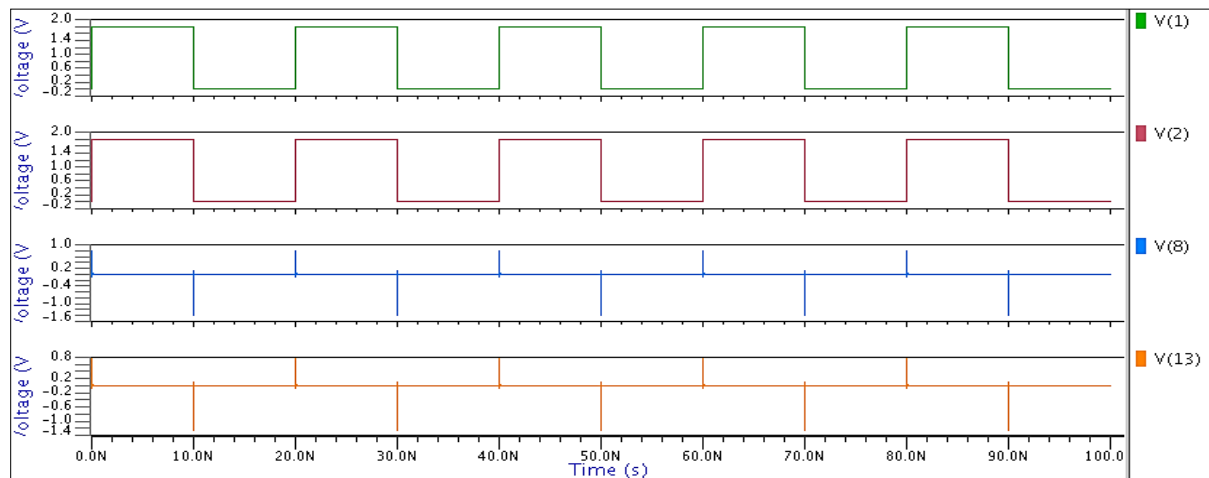


Fig. 8-A: High Speed PFD at 50 MHz (lock condition).

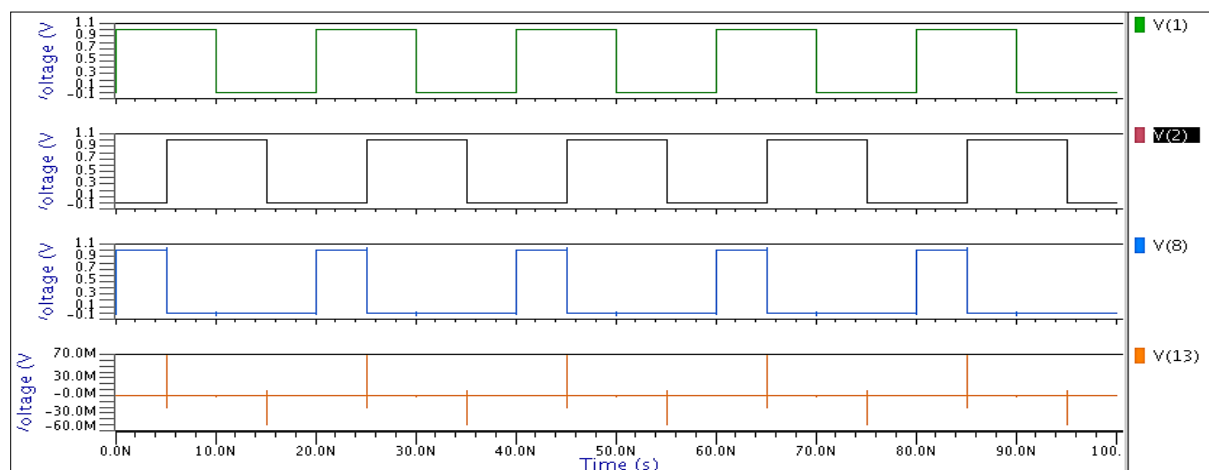


Fig. 8-B: High Speed PFD at 50 MHz (CLKREF Lead Condition).

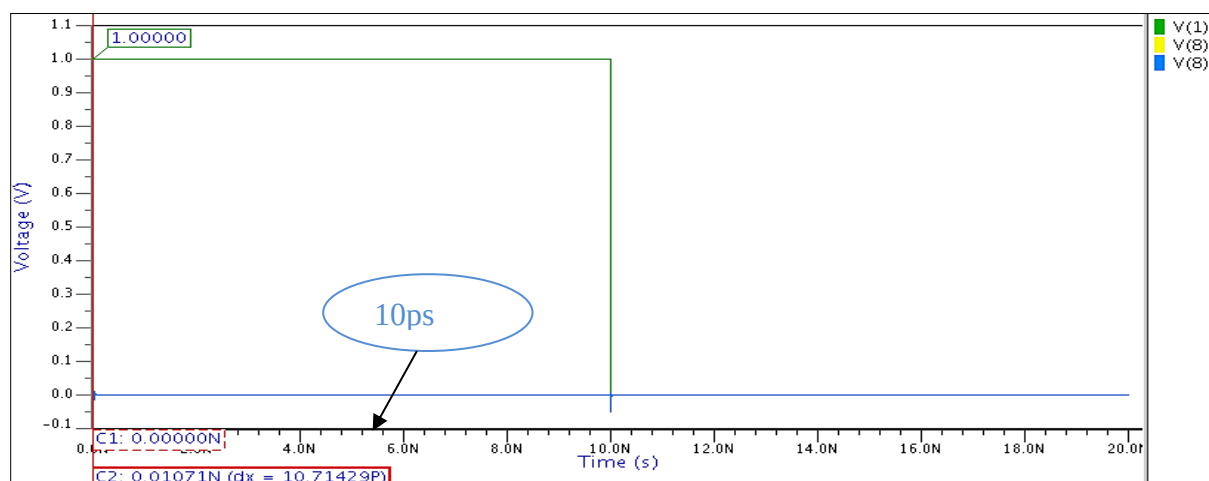


Fig. 8-C: Dead Zone of High Speed PFD at 50 MHz (Lock Condition).

Dead zone is cause of small phase error, when both the input signals of PFD are very close to each other but the output signals are not able to generate this error. So it will create the dead zone problem. Due to this dead zone problem at the output of PLL circuits, the jitter will produce and the sensitivity of the PFD is also affected. As shown in Figure 8-C, the dead zone is 10.7 ps. Dead Zone may be possible to reduce further by improvement in the circuit and design as well.

SIMULATION RESULTS IN 0.35 μm TECHNOLOGY

The result of the high speed PFD with CMOS 0.35 μm technology is as shown in Figure 9 A-B-C. Here the dead zone achieved upto 13.39 ps. Figure 10 represents the dead zone at 1 GHz frequency up to 2 ps in locked condition.

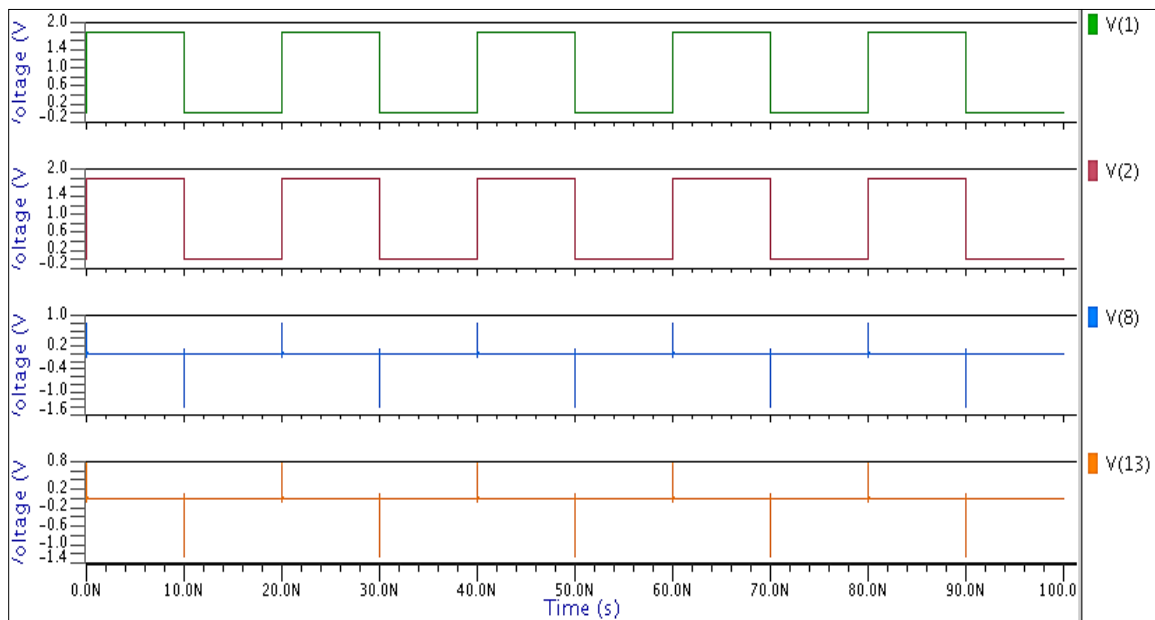


Fig. 9-A: High Speed PFD at 50 MHz (lock condition).

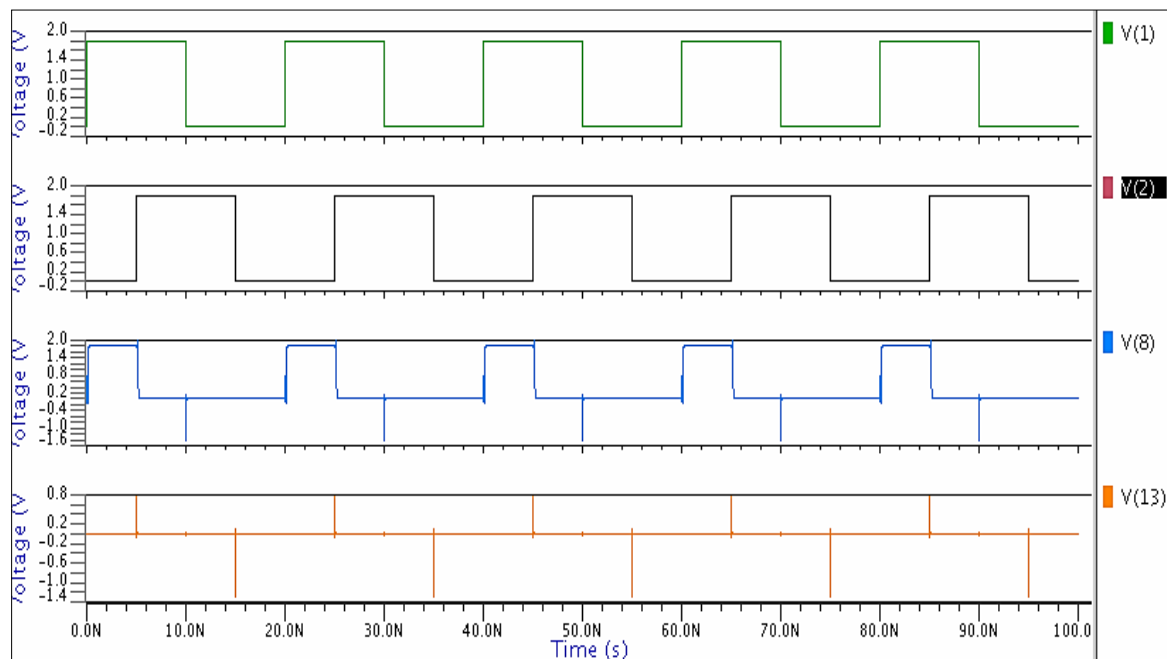


Fig. 9-B: High Speed PFD at 50 MHz (CLKREF lead condition).

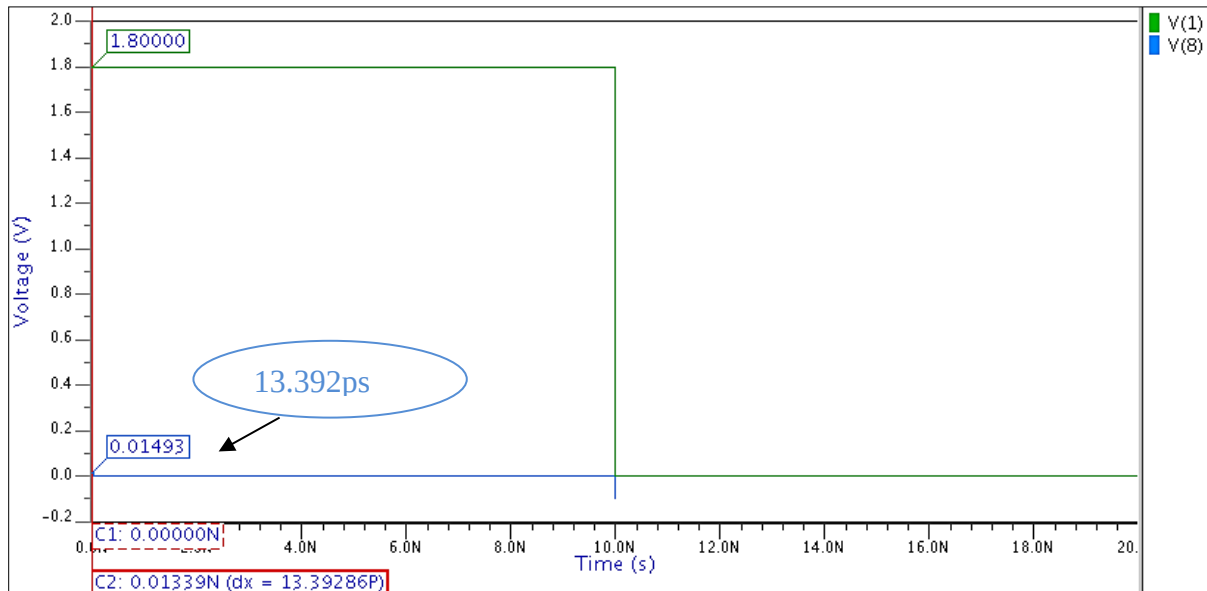


Fig.9-C: Dead zone of High Speed PFD at 50 MHz (lock condition).

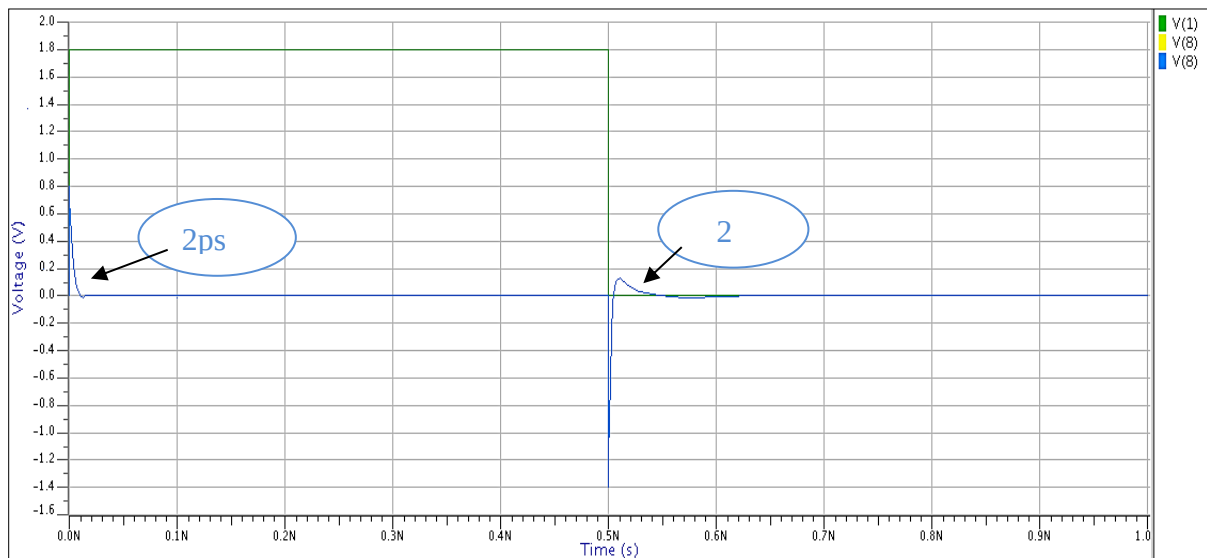


Fig.10: Dead zone of High Speed PFD at 1 GHz (lock condition).

Table 2 & 3 represents the comparative analysis of the simulation results with the references in 0.35 μm CMOS technology with 50 MHz and 1 GHz frequency inputs. The simulation results of 90 nm and 0.35 μm CMOS technology of High Speed Phase Frequency Detector circuit also compared with

different frequency inputs as shown in Table 4 & 5. As the frequency is increased, the dead zone can be reduced to a certain level. So overall jitter and power dissipation can be reduced. This type of High Speed Phase Frequency Detector circuit may be more preferable in Phase Locked Loop design.

Table 2: Design Specification of High Speed PFD at 50 MHz Frequency.

Parameter	Specification[11,12,13]	Design Results
Technology	0.35 μm	0.35 μm
Input frequency	50 MHz	50 MHz
Transistor count	18	18
Power Consumption	0.87-09 WATT	0.34-09 WATT
Dead-zone	14 ps	13 ps

Table 3: Design Specification of High Speed PFD at 1 GHz Frequency.

Parameter	Specification[11,12,13]	Design Results
Technology	0.35 μm	0.35 μm
Input frequency	1 GHz	1 GHz
Transistor count	18	18
Power Consumption	0.39E-09 WATT	0.423E- 09WATT
Dead-zone	2 ps	2 ps

Table 4: Simulation Result of High Speed PFD in 0.09 μm Technology.

Parameter	High Speed PFD	High Speed PFD	High Speed PFD
Technology	0.09 μm	0.09 μm	0.09 μm
Input Frequency	50 MHz	500 MHz	1 GHz
Transistor count	18	18	18
Power Consumption	0.26E-10 WATTS	0.83E-10 WATTS	0.97E-10 WATTS
Dead-zone	10 ps	4 ps	2 ps

Table 5: Simulation Result of High Speed PFD in 0.35 μm Technology.

Parameter	High Speed PFD	High Speed PFD	High Speed PFD
Technology	0.35 μm	0.35 μm	0.35 μm
Input Frequency	50 MHz	500 MHz	1 GHz
Transistor count	18	18	18
Power Consumption	0.34E-09 WATTS	0.42E-09 WATTS	0.42E-09 WATTS
Dead-zone	13 ps	5 ps	2 ps

CONCLUSION

In this paper Phase Frequency Detector, suitable for high speed applications, has been characterized and simulated. The Phase Frequency Detector is designed in TSMC0.35 μm and in Generic 90 nm technologies.

The Phase Frequency Detector is characterized in terms of the dead zone and power consumption. The simulation results allow the circuit designer to fully explore the tradeoffs/trade-offs in Phase Frequency Detector design, such as dead zone and power consumption.

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